Black-Box Dense Retrieval and Institutional Hardware Efficiency.

Izacard et al. (2022) demonstrated with Atlas that offloading factual storage to a dense retrieval index enables an 11B model to outperform an unguided 540B model on knowledge-intensive tasks. Shi et al. (2023) introduced REPLUG to prepend retrieved passages directly to frozen black-box language models without costly weight retraining.

Mao et al. (2024) developed FIT-RAG to prune irrelevant prompt tokens and cut memory consumption in half while improving factual accuracy on constrained hardware. Peng et al. (2025) noted that constructing entity-relationship knowledge graphs in GraphRAG incurs heavy LLM token and indexing latency compared to flat dense vector indexing.